

## V8.19.25.1

### Overview

This is a program or series of programs that work with each other to trade stocks. It consists of several modules to define rules, ingest data, send buy/sell commands and monitor open positions. It must be easy to modify a single modules code or logic without having to make significant changes to other modules. A primary concern is application speed. The application must evaluate the rules and data as fast a possible and send appropriate buy/sell commands to the broker module; delays of even a few milliseconds will have a negative impact.

As of V8.19.25.1, the following are not yet defined:

- programming Language (some 3<sup>rd</sup> parties use rust, java or python)
- when does the primary module write data to db
- what data needs to be stored in db, and how long to store it
- how will the modules work/communicate with each other
- how to parse the data from Polygon and broker so the primary module can make decisions

### Primary module

- Evaluates buy/sell conditions based on rules module and data from data module
- sends buy/sell to broker module
- Ingests data from broker module
- ingests data from data module
- ingests data from rules module
- writes data to psql
- If a database and it's tables, functions, triggers, etc, do not exist, it creates the database upon execution
- If a database exists but needs to be modified to support new functionality, modify DB accordingly (such as new tables, rows, triggers, etc)

### Data Module

- Data feed will be provided by <https://polygon.io>
- This module will consume data from the REST and Websocket API provided by Polygon
- will ingest data every 60 seconds from TopMarketMovers endpoint as described in <https://polygon.io/docs/rest/stocks/snapshots/top-market-movers>
- for all tickers listed in the top-market-movers, get the most recent news events as described in <https://polygon.io/docs/rest/stocks/news>
- when a trade is active, subscribe to the trades endpoint as described in <https://polygon.io/docs/websocket/stocks/trades>
- when a trade is active, subscribe to the quotes endpoint as described in <https://polygon.io/docs/websocket/stocks/quotes>
- when a trade is active, subscribe to the aggregates endpoint as described in <https://polygon.io/docs/websocket/stocks/aggregates-per-second>
- Parses/formats or performs ETL action to make data usable to the Primary Module. This may not be necessary, the format provided by polygon.io may be sufficient
- track account broker account value (investigate if we can can we get this from the broker module)
- track fees and charges from the broker for each order

### Broker Module

- will use LightSpeed Connect to execute and monitor trades as described in <https://lightspeed.com/trading/api-trading>
- monitors execution details
- defines order details such as quantity, limit, market, stop, equity route (AMEX, smart-ls, ARCA, etc), cancellation, etc.  
[https://d31x4u3ydvpo.cloudfront.net/manuals/Lightspeed\\_Connect\\_Getting\\_Started\\_Guide\\_8\\_14\\_24.pdf](https://d31x4u3ydvpo.cloudfront.net/manuals/Lightspeed_Connect_Getting_Started_Guide_8_14_24.pdf)

### Rules Module

- webpage where user inputs rules criteria
- criteria
  - Technical indicator = MACD, RSI, EMA, SMA
  - min/max price
  - volume
  - portfolio starting amount
  - max percent of portfolio or dollar value that can be invested
  - define risk reward ratio
  - max number of shares for an order